

**Saisaran M 251801281-2025-AIDS** ▾**S2****Started on** Thursday, 6 August 2026, 10:15 AM**State** Finished**Completed on** Tuesday, 25 August 2026, 12:54 PM**Time taken** 19 days 2 hours**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      int count = 0;
6
7      scanf("%d", &n);
8
9      int i = 1;
10     count++;
11
12     int s = 1;
13     count++;
14
15     while (count++, s <= n) {
16         i++;
17         count++;
18
19         s += i;
20         count++;
21     }
22
23     printf("%d\n", count);
24
25     return 0;
26 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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**Saisaran M 251801281-2025-AIDS** ▾**S2****Started on** Thursday, 6 August 2026, 10:54 AM**State** Finished**Completed on** Tuesday, 25 August 2026, 12:15 PM**Time taken** 19 days 1 hour**Marks** 0.00/1.00**Grade** 0.00 out of 10.00 (0%)

Question 1 | Incorrect Mark 0.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  // Global counter variable
4  int count = 0;
5
6  void func(int n) {
7      count++; // 1 step for checking if(n==1)
8      if (n == 1) {
9          count++; // 1 step for printf
10         printf("*");
11     }
12     else {
13         count++; // 1 step for initialization (i=1)
14         for (int i = 1; i <= n; i++) {
15             count++; // 1 step for condition check (i<=n) when true
16
17             count++; // 1 step for initialization (j=1)
18             for (int j = 1; j <= n; j++) {
19                 count++; // 1 step for condition check (j<=n) when true
20
21                 count++; // 1 step for printf("*")
22                 printf("*");
23
24                 count++; // 1 step for printf("*")
25                 printf("*");
26
27                 count++; // 1 step for break statement
28                 break;
29
30                 // Note: j++ and the false condition of j<=n are never reached
31                 // because of the break statement.
32             }
33             count++; // 1 step for increment (i++)
34         }
35         count++; // 1 step for condition check (i<=n) when false
```

```
36     }
37 }
38
39 int main() {
40     int n;
41     // No counter increments for declarations or scanf
42     scanf("%d", &n);
43
44     func(n);
45
46     // No counter increments for count variable printf
47     printf("\nCounter: %d\n", count);
48
49     return 0;
50 }
```

	Input	Expected	Got
✖	2	12	**** Counter: 17
✖	1000	5002	***** Counter: 7003

Some hidden test cases failed, too.

Your code must pass all tests to earn any marks. Try again.

Show differences

Incorrect

Marks for this submission: 0.00/1.00.

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**Saisaran M 251801281-2025-AIDS** ▾**S2****Started on** Thursday, 6 August 2026, 11:04 AM**State** Finished**Completed on** Tuesday, 25 August 2026, 12:16 PM**Time taken** 19 days 1 hour**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int count = 0;
4
5  void Factor(int num) {
6      count++;
7      for (int i = 1; i <= num; ++i) {
8
9          count++;
10         if (num % i == 0) {
11             count++;
12         }
13
14         count++;
15     }
16 }
17
18 int main() {
19     int num;
20
21     scanf("%d", &num);
22
23     Factor(num);
24
25     printf("%d", count);
26
27     return 0;
28 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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[Saisaran M 251801281-2025-AIDS](#) ▾**S2****Started on** Thursday, 6 August 2026, 11:12 AM**State** Finished**Completed on** Tuesday, 25 August 2026, 12:53 PM**Time taken** 19 days 1 hour**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int count = 0;
4  void function(int n) {
5      count++;
6      int c = 0;
7
8      count++;
9      for (int i = n / 2; i < n; i++) {
10
11          count++;
12          for (int j = 1; j < n; j = 2 * j) {
13
14              count++;
15              for (int k = 1; k < n; k = k * 2) {
16
17                  count++;
18                  c++;
19
20                  count++;
21              }
22
23              count++;
24          }
25
26          count++;
27      }
28  }
29
30  int main() {
31      int n;
32
33
34      scanf("%d", &n);
35
36      function(n);
37
38
39      printf("%d", count);
40
41      return 0;
42  }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Saisaran M 251801281-2025-AIDS ▾

S2

Started on	Thursday, 6 August 2026, 11:22 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 11:50 AM
Time taken	19 days
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int count = 0;
4
5  void reverse(int n) {
6      count++;
7      int rev = 0, remainder;
8
9
10     while (count++, n != 0) {
11
12         count++;
13         remainder = n % 10;
14
15         count++;
16         rev = rev * 10 + remainder;
17
18         count++;
19         n /= 10;
20     }
21
22     count++;
23 }
24
25 int main() {
26     int n;
27
28     scanf("%d", &n);
29
30     reverse(n);
31
32
33     printf("%d", count);
34
35     return 0;
36 }
37
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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